Paleomagnetic results from the Nain anorthosite and their tectonic implications

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As part of studies of anorthosites from the northern hemisphere, the paleomagnetism of the Nain anorthosite from Labrador is described. A total of 113 samples were collected from 21 sites comprising both dark and pale facies types. Although AF (alternating field) demagnetization was not useful in obtaining a stable component of magnetization, it was possible to isolate two oppositely directed (easterly and westerly) magnetizations through thermal demagnetization. Quite often, both magnetizations are found in the same specimen. On the basis of thermal demagnetization, the 18 sites of dark facies anorthosite are found to possess stable remanence at $D = 278.2°$, $I = 15.8°$, $k = 174$, $\alpha_{95} = 2.6°$ with the corresponding paleomagnetic pole of mixed polarity at 11°N, 149°W (δ_p, $\delta_m = 1.5°$, 3°). A magnetization with $D = 275.3°$, $I = 27.2°$, $k = 92$, $\alpha_{95} = 12.9°$ with the corresponding paleomagnetic pole at 15°N, 148°W (δ_p, $\delta_m = 7\frac{1}{2}°$, 14°) was isolated for the pale facies anorthosite. Comparison of the Nain results with those of other plutons from Labrador suggests a significant difference in pole positions for the various anorthosite bodies north of the Grenville Front, which may represent a slight north–south polar shift during the time interval of intrusion of these bodies.

Comme complément aux études sur les anorthosites dans l'hémisphère nord, on décrit ici le paléomagnétisme des anorthosites de Nain, dans le Labrador. On a recueilli un total de 113 échantillons dans 21 localités pour les types à faciès foncé et à faciès pâle. Bien que la désaimantation AF n'ait pas été utile pour définir une composante d'aimantation stable, il a été possible d'isoler deux aimantations de directions opposées (est et ouest) par désaimantation thermique. Assez souvent, on retrouve les deux aimantations dans le même échantillon. En se basant sur la désaimantation thermique, les 18 sites de l'anorthosite à faciès foncé possèdent une rémanence stable à $D = 278.2°$, $I = 15.8°$, $k = 174$, $\alpha_{95} = 2.6°$ avec des pôles paléomagnétiques correspondant et de polarité mixte à 11°N, 149°O (δ_p, $\delta_m = 1.5°$, 3°). Une aimantation avec $D = 275.3°$, $I = 27.2°$, $k = 92$, $\alpha_{95} = 12.9°$ avec un pôle correspondant à 15°N, 148°O (δ_p, $\delta_m = 7\frac{1}{2}$, 14°) a été isolée dans l'anorthosite à faciès pâle. La comparaison des résultats de Nain avec ceux d'autres plutons du Labrador suggère une différence significative dans la position des pôles pour les différentes masses d'anorthosite au nord du front de Grenville, ce qui peut s'expliquer par une faible dérive polaire nord–sud durant l'intervalle de temps d'intrusion de ces masses.

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Anorthosites and Their Paleomagnetism

Anorthosites and their associated intrusive rocks form the only deep-seated intrusive bodies of batholithic proportions apart from those of the granite kindred. Anorthositic rocks include the massif-type plutons in areas of high grade gneiss and schist and the layered irregular masses in mineralogically stratified intrusions. Unlike the granitic rocks and the stratified intrusions, the massif-type anorthosites occupy a special place in the crustal development during Precambrian times. Herz (1969) has summarized the relevant material pointing to the uniqueness in time and space of these plutonic bodies. (1) The anorthosites appear to lie within broad belts when plotted on a reconstruction of the continents before drifting: one in the present northern hemisphere (Fig. 1) and another in the southern hemisphere. The belt in the present northern hemisphere begins in the Ukraine and continues west through Sweden, southern Norway, the Outer Hebrides, Greenland, eastern Canada and the United States. (2) The apparent ages of the anorthosites cluster about 1300 ± 200 Ma. There is growing evidence (Isachsen 1969) that the anorthosites are the product of a unique worldwide event, perhaps occurring under thermal conditions in the upper layers of the Earth that have not been duplicated since the Precambrian. A comprehensive study of these bodies has been started with the hope that comparison of paleomagnetic poles of these massifs from land masses bordering the North Atlantic with excellent time correlations, will provide a powerful method of reconstructing the relative positions of these land masses.

The anorthositic bodies from within the Grenville Province of the Canadian Shield are in a special category in that they all yield typical Grenville ages, probably representing the age of the overprint of Grenville metamorphism on these bodies. The paleomagnetism of these plutons has been well

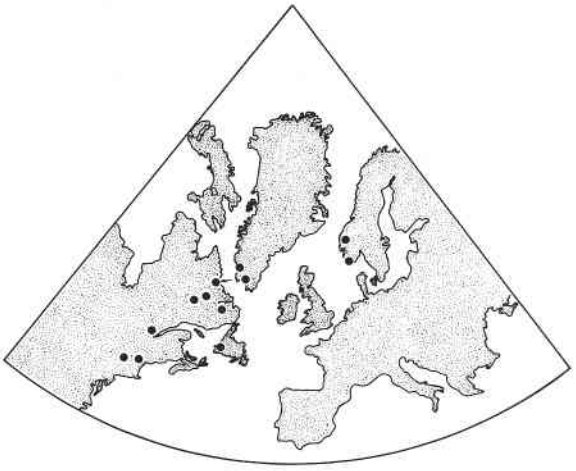


FIG. 1. Anorthosites of the northern hemisphere plotted on a North Atlantic reconstruction. Anorthosite locations are shown by solid dots. Arrow points to the Nain anorthosite for which paleomagnetic results are presented in this paper.

studied (Hargraves and Burt 1967; Palmer and Carmichael 1973; Fahrig *et al.* 1974; Hargraves and Roy 1974; Irving *et al.* 1974*b*; Ueno *et al.* 1975) and the results form the basis for constructing a polar wander path for the Grenville segment of the Canadian Shield (Irving *et al.* 1974*a*; Buchan and Dunlop 1976).

Anorthosites north of the Grenville Front were unaffected by the Grenville orogeny and are extremely fresh for Precambrian rock units. Anorthosite massifs at Michikamau and Harp Lakes and the coastal exposures around Nain and the related numerous adamellite suites fall in this category. The tectonic setting of these large intrusive bodies in Labrador is by no means clear. Emslie *et al.* (1972) reviewed the tectonic evidence and potassium–argon age data available for these bodies. According to these authors, the intrusions postdate the deformation and metamorphism associated with the Hudsonian orogeny (about 1700 Ma). K–Ar mineral ages around 1400 Ma suggest that the intrusions represent a completely separate igneous event unrelated to any recognized tectonism.

Anorthosites are proved to be excellent carriers of extremely stable magnetization residing in thin oriented rods lying within plagioclase crystals (Murthy *et al.* 1971; Palmer and Carmichael 1973; Irving *et al.* 1974*b*; Piper 1974; Emslie *et al.* 1976; Murthy and Rao 1976). The pole position for the Michikamau anorthosite was obtained by Murthy *et al.* (1968) and Emslie *et al.* (1976). The Harp Lake anorthosite was studied paleomagnetically by Irving *et al.* (1977). This paper describes the paleomagnetism of the other anorthosite body north of the Grenville Front, the massive exposures around Nain, Labrador. The paleomagnetism of the Nain anorthosite was studied with a two-fold purpose. Firstly, it is of interest to see whether all the anorthositic bodies north of the Grenville Front have similar magnetizations. Secondly, the anorthosites around Nain have been divided into three facies (see next section) and it is of interest to check the magnetic stability of the various facies. Dymek and Hargraves (1973) reported preliminary paleomagnetic results from this anorthosite body.

Geological Setting of the Nain Anorthosite

The Nain anorthosite in Labrador lies north of the Grenville Front and is unaffected by the Grenville orogeny. The anorthosites are marvellously well exposed in a deeply dissected skerry coastline with easy accessibility by water for distances of hundreds of kilometres. The area underlain by anorthosite and related intrusive rocks appears to be nearly 10 000 km^2. The Nain anorthosite had been under study for more than 40 years by the late E. P. Wheeler. The very detailed geological mapping by Morse and his colleagues (1972–1976) of this area resulted in the discovery of many intrusive bodies within the Nain anorthosite.

The Nain complex essentially consists of anorthosite and related basic rocks along with intermediate to acidic rocks collectively grouped by Wheeler (1960) into the adamellite series. Anorthosite in the strict sense probably forms less than half the mass of anorthositic rocks, the remainder being leuconorite to norite or more rarely leucotroctolite to troctolite. Wheeler (1960) subdivided the anorthosite complex into three major facies: dark, pale and buff-weathering. The dark facies is characterized by dark gray plagioclase with widespread presence of olivine and the lowest content of potassium of the three facies. The pale facies anorthosite is characterized by pale gray feldspar with hypersthene rather than olivine as the mafic mineral and has intermediate potassium content. Buff-weathering anorthosite is normally dominated by andesine and has higher potassium content than the other two. Rocks of the ademellite series were observed frequently to cross-cut anorthositic rocks suggesting a younger age for the adamellite series. The anorthosite complex is manifestly a complex of many smaller plutons emplaced over a period of time.

A reasonable number of isotopic dates are available for the Nain anorthosite and associated rocks. An early age reported by Morse (1964) suggests a minimum potassium–argon age of 1480 ± 50 Ma

for the Kiglapait layered intrusion and for the anorthosite, this value being comparable to the 1400 ± 80 Ma reported by Emslie (1964) for the Michikamau anorthosite. Within the Nain anorthosite complex, Barton (1974) did Rb–Sr studies on the minerals found in the lensoid bodies of coarsely crystalline plagioclase and clinopyroxene. The age of 1418 ± 25 Ma obtained from these studies is interpreted by Barton as the age of crystallization of the body and is probably a good approximation of the age of the surrounding anorthosite complex. Barton also obtained a K–Ar age for the Manvers granite, a fine to medium grained undeformed granite intruding the Kiglapait intrusion. His value of 1274 Ma is similar to the 1140 ± 40 Ma obtained by Beall *et al.* (1963) for the same body and sets an upper limit for the age of crystallization of the Kiglapait intrusion as 1274 Ma. Extensive literature is available (e.g. Morse 1969, 1972–1976) indicating that the Kiglapait layered intrusion is only slightly younger than the anorthosite massif to the south. K–Ar age determinations by the Geological Survey of Canada (Wanless *et al.* 1973) on the adamellite plutons yielded ages of around 1160 Ma that are interpreted as probably the age of the intrusion. Field evidence documented by Morse and his colleagues (Morse 1976) suggests that the adamellites post-date the anorthosite massif. Thus there is reasonable evidence that the Nain anorthosite massif has intruded at the same time as the Harp Lake and the Michikamau plutons approximately 1400 Ma ago.

Paleomagnetism of the Nain Anorthosite

Samples for the paleomagnetic study were collected during the 1973 field season at localities shown in Fig. 2, all from dark and pale facies anorthosites. Sampling was concentrated in three regions viz. Palungotok Island (7 sites, dark facies), a region south of Nain (11 sites, dark facies) and Paul Island (3 sites, pale facies). The region south of Nain includes localities described by Wheeler (1973) as containing giant pyroxene crystals. A total of 113 oriented hand samples was collected from 21 sites. Orientation was by both solar and magnetic compasses. Three to four cylinders, each about 2.5 cm in diameter and 2.4 cm in length, were obtained from each of the samples in the laboratory and their remanent magnetization was measured using a PAR SM-1 spinner magnetometer and an astatic magnetometer. The average magnetization of at least two cylinders was used as the basic statistical unit in the analysis. Measurements of magnetic susceptibility k were made using a susceptibility bridge described by Pätzold (1972). The

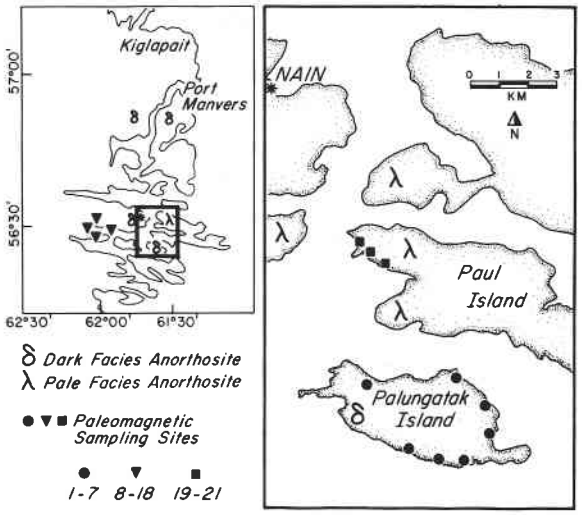


FIG. 2. Distribution of dark and pale facies anorthosites in the Nain area after Wheeler (1969).

intensities of natural remanent magnetization (NRM) are typically of the order of 10^{-4} emu/cm^3 (10^{-1} A/m) and susceptibilities range from 10^{-5} to 10^{-3} cgs units ($\approx 10^{-6}$ to 10^{-4} SI units). Values of NRM intensity, susceptibility and Koenigsberger ratio are given in the form of histograms in Fig. 3. It is of interest to note that although the Koenigsberger ratios are of reasonable value for a Precambrian rock unit, the values are substantially lower than the typical values of around 100 noted for the Egersund anorthosite of southern Norway (Murthy and Deutsch 1974). Although Fig. 3 does not differentiate between dark and pale facies magnetizations, sites of pale facies anorthosites are characterized by low remanence intensity, low susceptibility and low Koenigsberger ratios.

The mean directions of natural remanence of each site are plotted in Fig. 4. Although most site mean magnetizations are close to the present Earth's field direction, two sites from Palungotok Island have easterly declination with shallow negative inclinations. Grouping around the present Earth's field is probably indicative of viscous remanence acquired during Recent times. Alternating field (AF) demagnetization was carried out on pilot specimens in steps of 50 Oe (39.8×10^2 A/m). Response to AF demagnetization is of a complex nature as indicated for a typical specimen (N55) in Fig. 5a. Intensity tended to increase with increase in demagnetizing field in an oscillatory fashion and the corresponding variations in directions also seem to have an oscillatory character. After demagnetization in 1200 Oe (95.5 kA/m), the specimen has remanence at least as strong as the original NRM and the directional response seems to be of

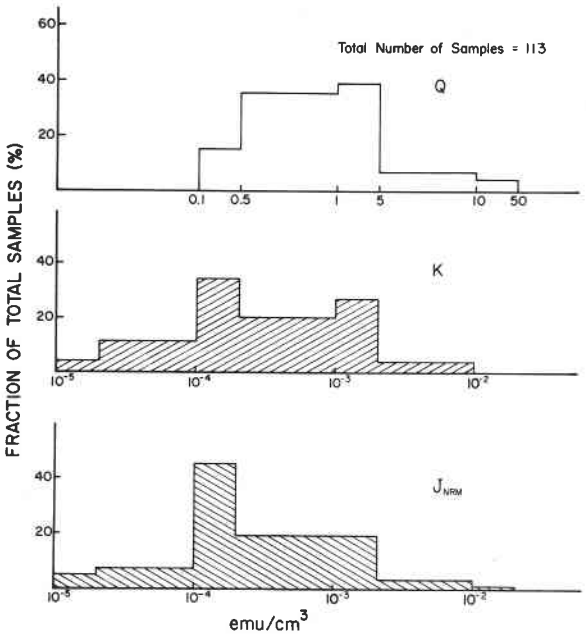


FIG. 3. Histograms of NRM intensity (J_{NRM}), magnetic susceptibility (k) and Koenigsberger ratio (Q) for the anorthosite samples collected.

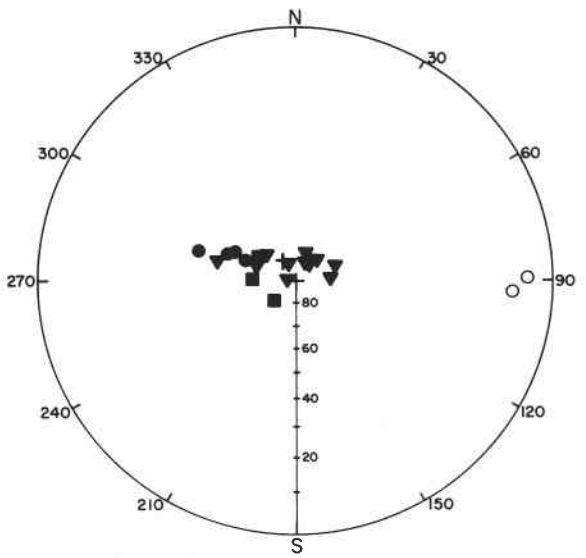


FIG. 4. NRM site mean directions of Nain anorthosite: ●, Palungotok Island; ▼, sites south of Nain; ■, pale facies anorthosite, Paul Island. Solid (open) symbols indicate positive (negative) inclinations. +, 1973 geomagnetic field at sampling locality.

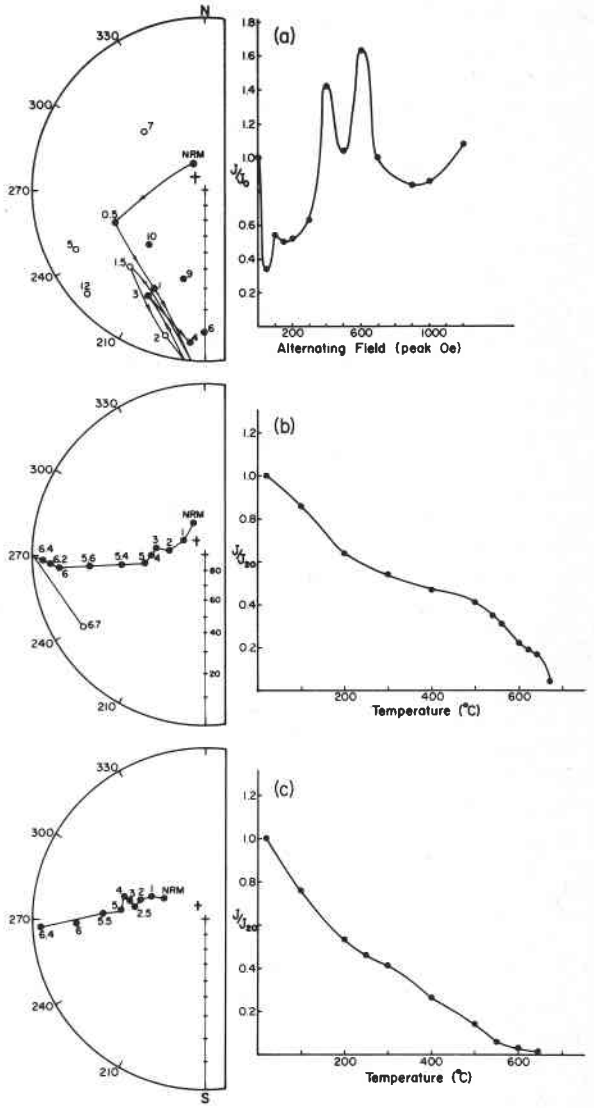


FIG. 5. Response of specimen N55 (dark facies anorthosite) to (a) alternating field demagnetization, (b) step-wise thermal demagnetization and (c) continuous thermal demagnetization. Directional response is plotted using Wulff's stereographic projection. Solid (open) symbols indicate positive (negative) inclinations. The labels by the points indicate in (a) the appropriate peak demagnetizing field in hundreds of oersteds (1Oe = 79.577 47 A/m) and in (b) and (c) the appropriate demagnetizing temperature in hundreds of degrees Celsius.

random nature. Such behavior is noted in most of the pilot specimens selected for AF demagnetization. At present, the author cannot give any reasonable explanation for the behavior of specimens during AF demagnetization, particularly in contrast to their behavior during thermal demagnetization (see Figs. 5b,c and later figures). Vector analysis of the AF demagnetization results was made to see whether a systematic magnetic vector had been erased (added) but no such vector was found. One possible explanation of the AF behavior may be the presence of two magnetizations of opposite polarities (easterly and westerly directed, see later results) in the same specimen car-

ried by one or more magnetic minerals with overlapping coercivity spectra. The presence of more than one component of magnetization with overlapping coercivities and equal magnitudes of remanent intensities and the erasure of some proportions of these at each demagnetization step may explain this behavior. But at this stage, such a model is hypothetical only and cannot be unequivocally proved.

In contrast to response to alternating fields, the specimens behaved in a more understandable fashion during thermal demagnetization. Both standard step-wise thermal demagnetization (where magnetizations are measured at room temperature after each demagnetization step) and continuous thermal demagnetization (where the magnetization of the specimen is measured at different temperatures during heating by use of an astatic magnetometer and furnace arrangement similar to the one described by Wilson (1962)) were employed. Results are shown in Figs. 5(b) and (c), respectively, for specimen N55. Specimen magnetizations moved to westerly declinations and shallow positive inclinations. Reasonably stable end points were also noticed. As thermal demagnetization is apparently more suitable for this rock unit, it was decided to use mostly thermal demagnetization for further analysis. Other examples of response of dark facies anorthosite to heating are shown in Figs. 6 and 7. Some specimens showed easterly declination and shallow negative inclinations (Fig. 6). It is possible also to notice both the easterly and westerly magnetizations in the same specimen (Fig. 7). However, it was found that it was difficult to isolate *both* magnetizations *accurately* in the *same* specimen. It seems either the easterly or westerly magnetization was dominant in any specimen (and site) and only one can be isolated accurately and the second magnetization may be noticed only as a trend. This behavior is very similar to that noticed by Irving *et al.* (1977) for the Harp Lake anorthosites. Based on the end points reached for any specimen, the remainder of the specimens from the site were thermally demagnetized at that temperature. The optimum temperatures of demagnetization varied from 500 to 640°C. It is of interest to note that all Palungotok Island sites became demagnetized in temperatures less than 570°C whereas for the other sites, temperatures in excess of 600°C are considered to be optimum temperatures (Table 1).

It was not possible to correlate either the easterly or westerly magnetization solely with hematite or magnetite. Sites 1 and 2 that showed only easterly magnetization became demagnetized in 530°C, indicative of magnetization carried by magnetite.

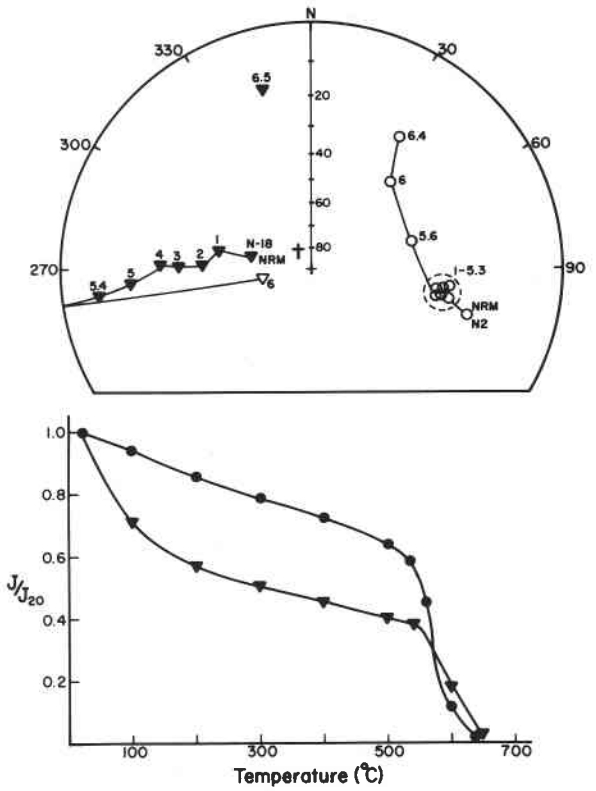


FIG. 6. Response of two specimens from Palungotok Island (dark facies) to thermal demagnetization. For explanation of labelling, see Fig. 5.

However, certain samples from sites south of Nain showed an easterly trending magnetization in temperatures in excess of 650°C, indicating that hematite in these specimens also carried an easterly magnetization. It may be concluded that both magnetite and hematite occur in the samples and the magnetization is either easterly or westerly.

Polished sections of anorthosite samples were prepared and studied using a Zeiss optical microscope at ×1000 magnification. Similar to findings in other anorthosites studied earlier, the opaques are in two forms, as large irregular grains of approximately 300 μm size and also as elongated rods about 50 μm long in the feldspars. Although the large irregular grains have been identified as magnetite, it was not possible to identify the very fine rods as magnetite or hematite. Possibly both hematite and magnetite occur in rod shapes and are responsible for the observed stable remanence.

The pale facies anorthosite is distinguished by the lack of large grains, but the fine rod-shaped grains are as numerous as in the dark facies anorthosite. The thermal demagnetization response of pale facies anorthosite samples is shown in Fig. 8.

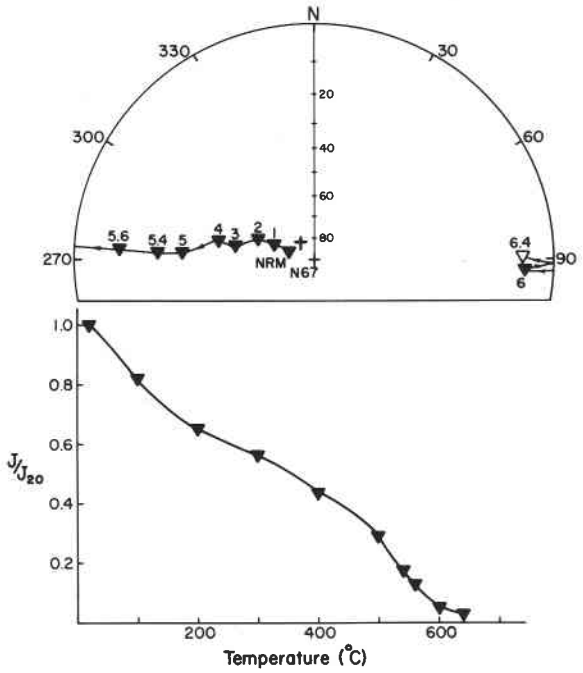


FIG. 7. Response of specimen N67 from sites south of Nain (dark facies) to thermal demagnetization. For explanation of labelling, see Fig. 5.

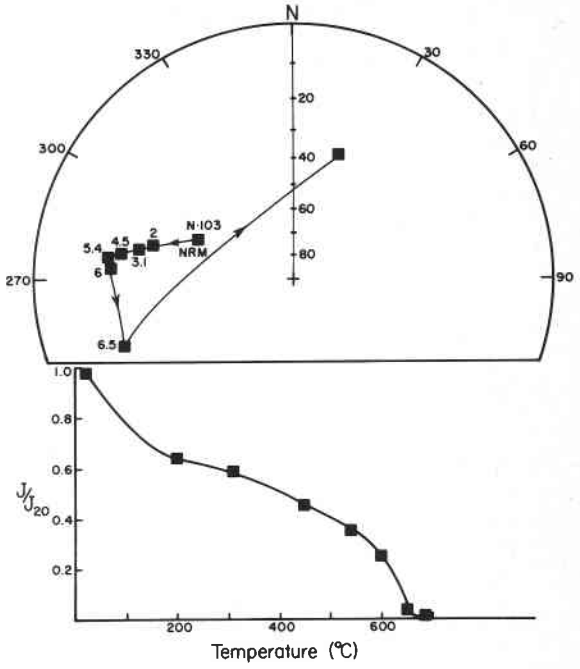


FIG. 8. Response of specimen N103 from Paul Island (pale facies) to thermal demagnetization. For explanation of labelling, see Fig. 5.

In general, the inclinations do not get as shallow as for the dark facies samples. No easterly magnetization was present in these sites.

In order to ascertain the nature of the magnetic mineral, powdered fractions of the samples were subjected to thermomagnetic analysis. The results are shown in Fig. 9. Magnetite is indicated as the magnetic mineral and the presence of hematite, if any, cannot be ascertained from thermomagnetic curves. This may be due to an instrumental difficulty in that fields of the order of only 1200 Oe (95.5 kA/m) are used in the experiments. In fields of that order, the presence of hematite coexisting with magnetite cannot be identified except when using a very sensitive Curie balance. Further, if the large-sized grains (about 300 μm) are magnetite, and hematite occurs only in the form of rods, then in a J–T experiment, hematite presence may not be readily detectable because of its low relative volume precentage in the sample, and low magnetic susceptibility.

In summary, both easterly and westerly magnetizations are present in the Nain anorthosite. Even within a site with predominantly one type of magnetization, some samples exhibit opposite magnetization (e.g. sites 13, 14). Quite often within a cylinder with dominant magnetization of one type, a trend is noticeable of magnetization of the second type. The mean magnetization directions for each site were computed irrespective of the sign of magnetization and are listed in Table 1. The overall mean directions of magnetization for each locality and the corresponding pole positions were computed and are presented in Table 2. Site mean directions after optimum thermal demagnetization are plotted in Fig. 10. Figure 11 shows the mean directions combined according to locality.

Discussion

The directions of magnetization of the Nain anorthosite seem to be slightly different from those in both the Michikamau and the Harp Lake anorthosites. The Nain anorthosite has a slightly steeper paleomagnetic inclination. Although this may be taken as an apparent indication of an incomplete removal of the secondary component at Nain, the facts that the easterly magnetized sites (e.g. sites 1, 2) actually moved to *steeper* negative inclinations on demagnetization and also that the easterly and westerly magnetizations differ almost exactly by 180° after demagnetization suggest that all the secondary magnetizations had been removed. In any paleomagnetic study of intrusives, it is useful to show evidence for the age of remanent magnetization to be the same as the age of the intrusion. In this study the following suggest it may be so: (1) the presence of a magnetic reversal and the almost 180°

TABLE 1. Paleomagnetic results of individual sites in the present study

Site	N^a	Natural remanent magnetization (NRM)						After thermal demagnetization					
		$\bar{D}^b$	$\bar{I}^c$	$\alpha_{95}{}^d$	K^e	R^f	T (°C)^g	$\bar{D}$	$\bar{I}$	α_{95}	K	R	$\delta_\omega{}^h$
Palungotok Island sites													
1	6	92.4	−9.2	15.6	19.3	5.741	530	95.6	−17.1	7.5	79.8	5.937	8.3
2	4	89.0	−5.6	29.4	10.8	3.721	530	94.2	−14.8	16.0	33.7	3.911	12.1
3	6	295.7	60.8	19.7	12.5	5.601	540	277.5	13.7	9.1	55.1	5.909	10.0
4	5	292.5	66.3	6.4	143	4.972	540	282.2	17.6	5.7	179	4.978	5.4
5	6	296.5	70.4	11.8	33.4	5.851	510	273.7	21.3	9.6	49.3	5.899	10.5
6	5	291.4	58.3	13.3	34.0	4.882	500	276.0	18.6	7.8	96.8	4.959	7.3
7	5	287.2	47.3	32.2	6.6	4.394	510	275.2	18.6	12.4	39.2	4.898	11.6
Sites south of Nain													
8	4	296.4	70.1	18.8	25.0	3.880	620	286.2	19.6	11.8	61.3	3.951	9.0
9	6	312.3	72.4	15.4	19.9	5.748	610	277.8	18.6	6.9	95.9	5.948	7.5
10	7	24.8	80.6	23.6	7.5	6.197	630	283.2	16.2	8.5	51.0	6.883	10.5
11	6	340.6	81.8	17.8	15.0	5.667	640	282.0	10.4	9.3	52.8	5.905	10.2
12	5	257.6	86.7	40.6	4.5	4.113	640	280.9	10.6	11.8	42.7	4.906	11.1
13	5	21.3	77.2	18.2	18.6	4.785	640	276.6	17.4	5.8	176	4.977	5.5
14	4	284.2	54.8	18.8	25.0	3.880	620	287.1	13.3	12.0	59.8	3.950	9.1
15	4	35.0	80.4	33.4	8.5	3.648	530,600	265.0	13.2	16.1	33.5	3.911	12.1
16	4	44.7	78.9	18.6	25.2	3.881	610	100.5	−9.1	9.5	95.0	3.968	7.2
17	4	69.7	70.4	25.2	14.2	3.789	600	99.5	−19.8	6.5	199	3.985	5.0
18	5	85.3	74.0	15.7	24.6	4.837	600	94.4	−14.1	11.2	47.7	4.916	10.5
Paul Island sites													
19	4	300.2	70.1	10.7	74.2	3.960	650	285.7	28.4	11.5	64.8	3.954	8.7
20	3	270.4	71.0	18.0	47.8	2.958	640	267.3	26.4	16.5	56.6	2.965	8.8
21	3	225.1	76.4	54.7	6.2	2.675	640	273.1	26.1	28.2	20.2	2.901	14.8

[a] N, number of samples.
[b] $\bar{D}$, mean declination, degrees east of north.
[c] $\bar{I}$, mean inclination, degrees positive downwards.
[d] α_{95}, radius of 95% circle of confidence, in degrees.
[e] K, Fisher's precision parameter.
[f] R, length of resultant of N vectors.
[g] T, optimum temperature of demagnetization, in degrees Celsius.
[h] δ_ω, within-site standard deviation.

difference between easterly and westerly magnetizations, (2) the presence of both hematite and magnetite and the fact that the two magnetic minerals were found to carry both easterly and westerly magnetizations, (3) the large divergence of cleaned remanence directions from that of the present Earth's field at the sampling locality and (4) the unmetamorphosed state of these rocks.

The magnetization of Palungotok Island sites and sites south of Nain are virtually indistinguishable. Although the α_{95} circle for the Paul Island sites does intersect the α_{95} circles for the other two localities, a simple F-test reveals that magnetization of Paul Island sites is significantly different from those of the other localities. It may be said that the pale facies anorthosite has a significantly different magnetization from the dark facies anorthosite. The different lithology is compatible with such a possibility. Since their directions lie between the two polarity directions of the dark facies samples, an alternative explanation would be that a portion of their magnetization is due to the opposite polarity but that it cannot be isolated or removed from the dominant westerly downward component. However, as there is no corresponding opposite polarity direction isolated for pale facies samples, it would be more appropriate to discuss the results in terms of dark and pale facies anorthosites. The magnetizations are summarized in the last two rows of Table 2. The paleomagnetic poles obtained in the present study are plotted in Fig. 12 along with poles for anorthositic rocks from Labrador with polar track 4A of Irving and McGlynn (1976) relative to the Canadian Shield forming the background. The reader is referred to the reviews by Irving and McGlynn (1976) and by Stewart and Irving (1974) for the listing of other pole positions of similar age. Paleomagnetic poles for the Nain anorthosite fall north of the poles for Harp Lake and Michikamau anorthosites, although in very good general agreement with the pole positions for the 1400 Ma period. Also they are significantly north of the 'mean Elsonian' pole at 6°S, 146°W (A_{95} = 8°) calculated by Evans *et al.* (1975).

Because of the rather small α_{95} circles for the Nain, Harp Lake and Michikamau results, the difference in pole positions may represent a significant polar shift during the time interval of intrusion of

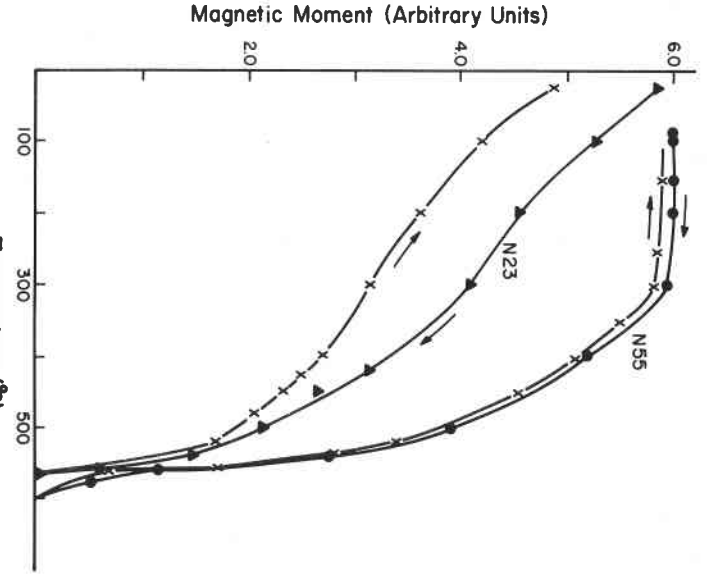


FIG. 9. Curie point determinations of powdered samples of dark facies anorthosite from Palungotok Island (N23) and sites south of Nain (N55).

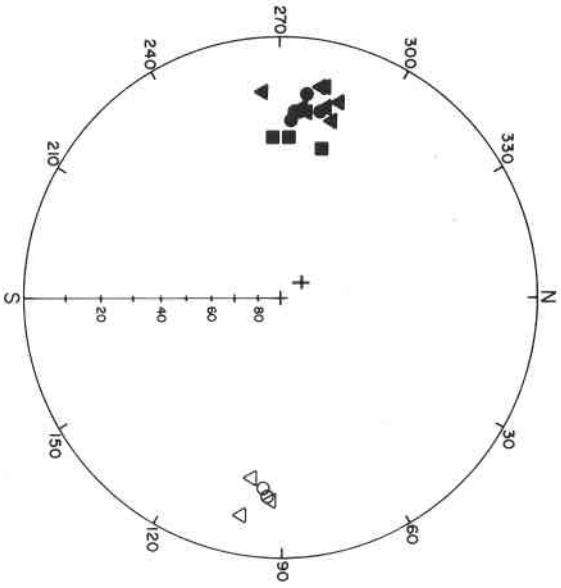


FIG. 10. Site mean directions of magnetization of Nain anorthosite, after optimum thermal demagnetization. ●, Palungotok Island; ▼, sites south of Nain; ■, Pale facies anorthosite, Paul Island. +, 1973 geomagnetic field at sampling locality.

these bodies. In order to explain the magnetization of the Harp dikes, Irving *et al.* (1977) suggested a possible double loop between polar tracks 4A and 3B. Such a double loop can accommodate the Nain

TABLE 2. Mean directions of magnetization and pole positions for Nain anorthosite

		Natural remanent magnetization (NRM)					After optimum thermal demagnetization								
Locality	N^a	$\bar{D}$	$\bar{I}$	α_{95}	K	R	$\bar{D}$	$\bar{I}$	α_{95}	K	R	$\delta_b{}^b$	λ'^c	ϕ'^d	$\delta_p,\delta_m{}^e$
Palungotok Island[f]	7	282.4	46.6	21.9	8.6	6.299	276.2	17.4	2.9	444	6.986	3.6	10.8°N	151.3°W	1.6, 3.0
Sites south of Nain[g]	11	350.9	82.4	9.0	26.6	10.624	279.4	14.8	4.0	133	10.925	6.7	11.4°N	147.8°W	2.1, 4.1
Paul Island	3	270.5	74.6	16.7	55.5	2.964	275.3	27.2	12.9	92.2	2.978	6.9	14.9°N	147.8°W	7.6, 14.0
All sites[h]	21	293.9	72.3	11.0	9.3	18.854	277.8	17.4	3.0	116	20.828	7.3	11.7°N	150.0°W	1.6, 3.1
Dark facies	18	297.5	71.6	12.9	8.2	15.917	278.2	15.8	2.6	174	17.902	6.0	11.2°N	149.1°W	1.4, 2.7
Pale facies	3	270.5	74.6	16.7	55.5	2.964	275.3	27.2	12.9	92.2	2.978	6.9	14.9°N	147.8°W	7.6, 14.0

[a] N, number of sites.
[b] δ_b, between-site standard deviation in degrees.
[c] λ', latitude of pole position.
[d] ϕ', longitude of pole position.
[e] δ_p, δ_m semi-angles of the oval of confidence about the pole position.
[f] Directions of sites 1 and 2 reversed in calculating mean direction.
[g] Directions of sites 16–18 reversed in calculating mean direction after optimum thermal demagnetization.
[h] Directions of sites 1, 2 and 16–18 reversed in calculating mean direction after optimum thermal demagnetization.

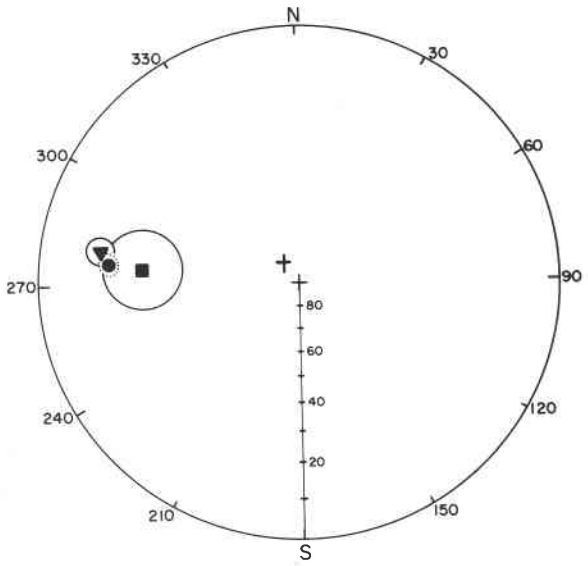


FIG. 11. Mean directions of magnetization for ●, Palungotok Island; ▼, sites south of Nain and ■ pale facies anorthosite, Paul Island. +, 1973 geomagnetic field at sampling locality. Solid circles around the mean directions are circles of 95% confidence around the mean.

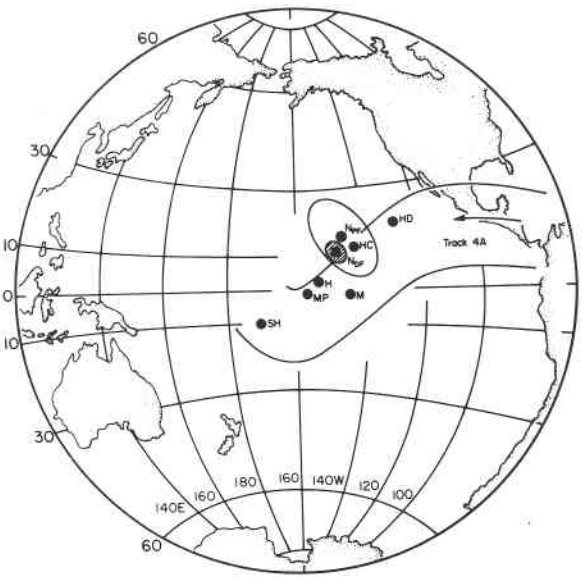


FIG. 12. Paleomagnetic pole positions for the Nain anorthosite (dark and pale facies) and other 1400 Ma plutons from Labrador plotted with Irving and McGlynn's polar track 4A as background. N_{DF}, Nain, dark facies; N_{PF}, Nain, pale facies; H, Harp Lake anorthosite; HC, Harp country rocks; HD, Harp dikes; M, Michikamau anorthosite; MP, Mistastin pluton; SH, Shabagamo gabbro. Shaded portions are ovals of confidence around the Nain results.

results. Alternatively, one may prefer not to invoke subordinate loops at this stage of availability of data. In such a case, the Nain results suggest that, although the predominant trend of track 4A is east–west, there may be a significant north–south apparent polar shift around the time of intrusion of the anorthosites.

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